

## PRANAV Khera

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### SUMMARY

An aspiring Engineer and Data Scientist pursuing a dual-degree path in Information Technology and AI. Experienced in building data-driven applications, from 3D web visualizations to predictive machine learning models. Proven ability to integrate AI APIs into full-stack projects and a background in competitive technical events.

### EDUCATION

**IIT Guwahati** | B.Sc. in Data Science and Artificial Intelligence | Expected 2029 **Guru Tegh Bahadur Institute of Technology** | B.Tech in Information Technology | Expected 2029 **CBSE (Class XII)** | Completed

### TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, Scikit-learn), C, JavaScript (Next.js, React).
- **Machine Learning:** Regression, Classification, Clustering, Dimensionality Reduction, Isolation Forest.
- **Data Science:** Exploratory Data Analysis (EDA), Model Deployment, Data Visualization.
- **Web Development:** React Three Fiber, Firebase, Supabase, Tailwind CSS, FastAPI, Flask.
- **Tools:** Git/GitHub, Gemini API, FxSound Optimization.

### KEY PROJECTS

**Fraud-Detector Engine** | *Python (FastAPI, NetworkX, SHAP), Next.js, Tailwind CSS, Gemini API*

- Developed an advanced fraud detection engine focusing on real-time, graph-based anomaly detection and behavioral data analysis.
- Integrated the Gemini API to generate clear, natural language explanations for flagged fraudulent activities.
- Built a responsive full-stack interface to visualize explainable anomaly outputs using Next.js and Tailwind CSS.

**Payout-App (Startup-Investor Platform)** | *Next.js, Tailwind CSS, Firebase, Python, Flask*

- Built a secure platform for startups and investors featuring financial fraud prevention and secure milestone-based payments.
- Implemented an escrow system utilizing Firestore logic to safely manage and release funds based on project progress.
- Developed a Python/Flask microservice integrating an Isolation Forest machine learning model for AI-driven fraud detection.

#### **Stellaris** | *Next.js, React Three Fiber, Three.js*

- Developed a 3D interactive star map and constellation visualizer.
- Implemented 2D canvas pattern-matching for recognizing the 88 IAU constellations.
- Upgraded the UI to support immersive 3D navigation of celestial bodies.

#### **Macros Analyzer** | *Python, Gemini API, Firebase*

- Built an AI-powered application that tracks food macronutrients directly from user-uploaded photos.
- Integrated the Gemini API for image recognition and nutrient estimation.
- Managed real-time data storage and user authentication using Firebase.

#### **Loan Approval Prediction** | *Python, Scikit-learn, Pandas*

- Engineered a machine learning model to predict loan eligibility based on user demographics and financial history.
- Performed extensive data cleaning and feature engineering to handle missing values.
- Achieved a peak model accuracy of 82.29% using Logistic Regression.

### **ACHIEVEMENTS & EXTRACURRICULARS**

- **Capture The Flag (CTF):** Participant in cybersecurity challenges at Delhi Technological University (DTU).
- **Interests:** Enthusiastic about Desi Hip Hop and optimizing audio hardware performance.

### **LANGUAGES**

- English (Professional)
- Hindi (Native)
- Punjabi (Native)

